

ANTONIO CRUCIANI

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<https://antonio-cruciani.github.io/>

Short Bio

Antonio Cruciani is a postdoctoral researcher in Computer Science at Aalto University, working in Prof. Jukka Suomela's group. His research focuses on randomized algorithms for robust and scalable computation, with emphasis on distributed computing, graph mining, temporal graphs, and dynamic networks. He obtained his Ph.D. in Computer Science from the Gran Sasso Science Institute in 2025, with a dissertation on centralized and distributed algorithms for evolving graphs. His work has appeared in international venues including ECML-PKDD, ICDM, DISC, PODC, SIROCCO, ICDCN, and SEA. He has held visiting research positions at IIT Madras, KTH Royal Institute of Technology, the University of Hamburg, and the University of Padua.

Fields of Research Expertise

Distributed Computing, Graph Mining, Randomized Algorithms, Temporal Graphs, Data Mining.

Current Position

Postdoctoral Researcher, Aalto University, Computer Science. Research Group of Prof. [Jukka Suomela](#).

*Espoo, FI
Mar. 2025 – Present*

Education

Ph.D. in Computer Science, Gran Sasso Science Institute (GSSI).
Thesis: *Models and Algorithms for Temporal Betweenness Centrality and Dynamic Distributed Data Structures*. **Grade:** Excellent.
Supervisors: [Francesco Pasquale](#) (University of Rome “Tor Vergata”), [Pierluigi Crescenzi](#) (GSSI)
Reviewers: [Christian Scheideler](#) (UPB), [Fabio Vandin](#) (UNIPD).

*L'Aquila, IT
Mar. 2025*

M.S. in Computer Science, University of Rome “Tor Vergata”,
Thesis: *Dynamic Random Graphs and unstructured P2P networks, analysis of two models inspired by the Bitcoin network*. Summa cum Laude.
Supervisor: [Francesco Pasquale](#) (University of Rome “Tor Vergata”)

*Rome, IT
2020*

B.S. in Computer Science, University of Rome “Tor Vergata”.
Thesis: *Efficient learning methods for playlist prediction*.
Supervisor: [Giorgio Gambosi](#) (University of Rome “Tor Vergata”)

*Rome, IT
2017*

Visiting Research Periods

Visiting Research Fellow, Department of Computer Science and Engineering, University of Padua.
 Host Professor: [Leonardo Pellegrina](#)

*Padova, IT
February 2026*

Visiting Research Fellow, KTH Royal Institute of Technology.
 Host Professor: [Aristides Gionis](#)

*Stockholm, SE
June 2025*

Visiting Research Fellow, Department of Computer Science University of Hamburg.
 Host Researcher: [Thorsten Götte](#)

*Hamburg, DE
January 2025*

Visiting Research Fellow, Department of Computer Science and Engineering,

Chennai, IN

IIT Madras.
Host Professor: [John Augustine](#)

1st-Aug. – 31st-Oct. 2024

Visiting Research Fellow, Department of Computer Science and Engineering,
University of Padua.
Host Professor: [Leonardo Pellegrina](#)

Padova, IT
July 2024

Visiting Research Fellow, Department of Computer Science and Engineering,
IIT Madras.
Host Professor: [John Augustine](#)

Chennai, IN
1st-Aug. 2023 – 27th-Feb.
2024

Visiting Research Fellow, Big Data Analytics Lab, Fondazione Ugo Bordoni
Host Researcher: [Giambattista Amati](#)

Rome, IT
Feb. – Nov. 2020

Teaching Experience

Invited Teacher, CS-E4565 - *Combinatorics of Computation D.* (MSc), Aalto
University (lecture delivery and guided problem solving, office hours, grading).
Topics: probabilistic method, theory of concentration, expander graphs.

Finland, FI
April - May. 2026

Invited Teacher, CS-E4565 - *Combinatorics of Computation D.* (MSc), Aalto
University (lecture delivery, guided problem solving, office hours, grading). **Topics:**
introduction on probability, the probabilistic method, and derandomization across
four sessions.

Finland, FI
April - May. 2025

Teaching Assistant, Computability and Computational Complexity Theory (BSc),
University of Rome “Tor Vergata”. (lecture delivery and guided problem solving)
Topics: computability and computational complexity.

Rome, IT
Oct. 2018 – Jun. 2019

Teaching Assistant, Computer Programming in C (BSc), University of Rome
“Tor Vergata”. Led exercise sessions focused on problem solving and programming
practice, supported students during in-class exercises, and graded weekly homework
assignments.

Rome, IT
Oct. 2017 – Jun. 2018

Professional Experience

Software Developer, WeDot Rome.

Rome, IT
Oct. 2015 – Jun. 2016

Software Developer, New System.

Falerone, IT
Jun. – Sep. 2010

Languages

◦ English (fluent)

◦ Italian (mother tongue)

Publications

In case of theoretical computer science conferences, authors are sorted alphabetically, otherwise by contribution.

Selected Publications

1. **MANTRA: Temporal Betweenness Centrality Approximation Through Sampling.** ECML-PKDD 2024.
Reason for selection: This is one of my single-authored papers and one of the clearest examples of my independent research profile. It develops a progressive sampling framework for temporal betweenness centrality with rigorous approximation guarantees and extensive experimental validation.
2. **Fast Percolation Centrality Approximation with Importance Sampling.** ICDM 2025.
Reason for selection: This work extends my research on scalable graph mining and sampling-based approximation to percolation centrality. I selected it because it combines theoretical sample-complexity analysis with an efficient importance-sampling algorithm for large graphs.

3. **Highly Dynamic and Fully Distributed Data Structures.** DISC 2025.
Reason for selection: This paper represents my work on highly dynamic distributed networks and shows how to maintain distributed data structures under churn. I selected it because it connects my PhD research with my broader goal of designing robust protocols for evolving distributed systems.
4. **New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs.** DISC 2025.
Reason for selection: This work contributes to foundational distributed computing by showing limits on the advantage of quantum-LOCAL algorithms for a broad class of distributed linear programs. I selected it because it shows my ability to contribute to deep theoretical questions beyond my main line on dynamic networks.
5. **propagate: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs.** ECML-PKDD 2023.
Reason for selection: This paper was my first major contribution to scalable graph mining. I selected it because it introduced efficient sampling-based methods for estimating distance-based graph metrics and helped establish the research direction that later led to MANTRA and my work on centrality approximation.

Conference Proceedings

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- [1] **A. Cruciani**, A. Das, A. Raevskaya, and J. Suomela. “Brief Announcement: It Does Not Matter How You Define Locally Checkable Labelings”. In: *Proceedings of the ACM Symposium on Principles of Distributed Computing*. PODC ’26. Association for Computing Machinery, 2026. ISBN: 9798400725128. DOI: [10.1145/3796701.3815923](https://doi.org/10.1145/3796701.3815923). URL: <https://doi.org/10.1145/3796701.3815923>. Acceptance rate = 22%.
 - [2] **A. Cruciani**, A. Das, M. Equi, H. Lievonon, D. Luong-Le, A. Modanese, and J. Suomela. “Brief Announcement: Is a LOCAL Algorithm Computable?”. In: *Proceedings of the ACM Symposium on Principles of Distributed Computing*. PODC ’26. Association for Computing Machinery, 2026. ISBN: 9798400725128. DOI: [10.1145/3796701.3815908](https://doi.org/10.1145/3796701.3815908). URL: <https://doi.org/10.1145/3796701.3815908>. Acceptance rate = 22%.
 - [3] **A. Cruciani**. “Maintaining a Bounded Degree Expander in Dynamic Peer-to-Peer Networks”. In: *Structural Information and Communication Complexity - 33rd International Colloquium, SIROCCO 2026, Durham, UK, June 9-11, 2026, Proceedings*. Lecture Notes in Computer Science. Springer, 2026. DOI: [10.1007/978-3-032-26465-7_11](https://doi.org/10.1007/978-3-032-26465-7_11). URL: https://doi.org/10.1007/978-3-032-26465-7_11. Acceptance rate = 43%.
 - [4] **A. Cruciani** and L. Pellegrina. “Fast Percolation Centrality Approximation with Importance Sampling”. In: *IEEE International Conference on Data Mining, ICDM 2025, Washington DC, USA, November 12-15, 2025*. IEEE, 2025, pp. 1125–1134. DOI: [10.1109/ICDM65498.2025.00121](https://doi.org/10.1109/ICDM65498.2025.00121). URL: <https://doi.org/10.1109/ICDM65498.2025.00121>. Acceptance rate = 8.9%.
 - [5] J. Augustine, **A. Cruciani**, and I. A. Gillani. “Brief Announcement: Highly Dynamic and Fully Distributed Data Structures”. In: *39th International Symposium on Distributed Computing (DISC 2025)*. Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 47:1–47:7. ISBN: 978-3-95977-402-4. DOI: [10.4230/LIPIcs.DISC.2025.47](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.47). URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.47>. Acceptance rate = 30.5%.
 - [6] A. Balliu, C. Coupette, **A. Cruciani**, F. d’Amore, M. Equi, H. Lievonon, A. Modanese, D. Olivetti, and J. Suomela. “New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs”. In: *39th International Symposium on Distributed Computing (DISC 2025)*. Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 11:1–11:22. ISBN: 978-3-95977-402-4. DOI: [10.4230/LIPIcs.DISC.2025.11](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.11). URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.11>. Acceptance rate = 30.5%.
 - [7] **A. Cruciani**. “Brief Announcement: Maintaining a Bounded Degree Expander in Dynamic Peer-To-Peer Networks”. In: *39th International Symposium on Distributed Computing (DISC 2025)*. Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 53:1–53:7. ISBN: 978-3-95977-402-4. DOI: [10.4230/LIPIcs.DISC.2025.53](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.53). URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.53>. Acceptance rate = 30.5%.

- [8] **A. Cruciani**. “MANTRA: Temporal Betweenness Centrality Approximation Through Sampling”. In: *Machine Learning and Knowledge Discovery in Databases. Research Track - European Conference, ECML PKDD 2024, Vilnius, Lithuania, September 9-13, 2024, Proceedings, Part I*. Ed. by A. Bifet, J. Davis, T. Krilavicius, M. Kull, E. Ntoutsi, and I. Zliobaite. Vol. 14941. Lecture Notes in Computer Science. Springer, 2024, pp. 125–143. DOI: [10.1007/978-3-031-70341-6_8](https://doi.org/10.1007/978-3-031-70341-6_8). URL: https://doi.org/10.1007/978-3-031-70341-6_8. Acceptance rate = 24%.
- [9] G. Amati, **A. Cruciani**, D. Pasquini, P. Vocca, and S. Angelini. “propagate: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs”. In: *Machine Learning and Knowledge Discovery in Databases: Research Track - European Conference, ECML PKDD 2023, Turin, Italy, September 18-22, 2023, Proceedings, Part III*. Ed. by D. Koutra, C. Plant, M. G. Rodriguez, E. Baralis, and F. Bonchi. Vol. 14171. Lecture Notes in Computer Science. Springer, 2023, pp. 671–688. DOI: [10.1007/978-3-031-43418-1_40](https://doi.org/10.1007/978-3-031-43418-1_40). URL: https://doi.org/10.1007/978-3-031-43418-1_40. Acceptance rate = 24%.
- [10] R. Becker, P. Crescenzi, **A. Cruciani**, and B. Kodric. “Proxying Betweenness Centrality Rankings in Temporal Networks”. In: *21st International Symposium on Experimental Algorithms, SEA 2023, July 24-26, 2023, Barcelona, Spain*. Ed. by L. Georgiadis. Vol. 265. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023, 6:1–6:22. DOI: [10.4230/LIPIcs.SEA.2023.6](https://doi.org/10.4230/LIPIcs.SEA.2023.6). URL: <https://doi.org/10.4230/LIPIcs.SEA.2023.6>. Acceptance rate = 50%.
- [11] **A. Cruciani** and F. Pasquale. “Dynamic graph models inspired by the Bitcoin network-formation process”. In: *24th International Conference on Distributed Computing and Networking, ICDCN 2023, Kharagpur, India, January 4-7, 2023*. ACM, 2023, pp. 125–134. DOI: [10.1145/3571306.3571398](https://doi.org/10.1145/3571306.3571398). URL: <https://doi.org/10.1145/3571306.3571398>. Acceptance rate = 34%.
- [12] **A. Cruciani** and F. Pasquale. “Brief Announcement: Dynamic Graph Models for the Bitcoin P2P Network: Simulation Analysis for Expansion and Flooding Time”. In: *Stabilization, Safety, and Security of Distributed Systems - 24th International Symposium, SSS 2022, Clermont-Ferrand, France, November 15-17, 2022, Proceedings*. Ed. by S. Devismes, F. Petit, K. Altisen, G. A. D. Luna, and A. F. Anta. Vol. 13751. Lecture Notes in Computer Science. Springer, 2022, pp. 335–340. DOI: [10.1007/978-3-031-21017-4_23](https://doi.org/10.1007/978-3-031-21017-4_23). URL: https://doi.org/10.1007/978-3-031-21017-4_23.
- [13] G. Amati, S. Angelini, **A. Cruciani**, G. Fusco, G. Gaudino, D. Pasquini, and P. Vocca. “Topic Modeling by Community Detection Algorithms”. In: *OASIS@HT 2021: Proceedings of the 2021 Workshop on Open Challenges in Online Social Networks, Virtual Event, Ireland, 30 August 2021*. Ed. by B. Guidi, A. Michienzi, and L. Ricci. ACM, 2021, pp. 15–20. DOI: [10.1145/3472720.3483622](https://doi.org/10.1145/3472720.3483622). URL: <https://doi.org/10.1145/3472720.3483622>.
- [14] **A. Cruciani**, D. Pasquini, G. Amati, and P. Vocca. “About Graph Index Compression Techniques”. In: *Proceedings of the 10th Italian Information Retrieval Workshop, Padova, Italy, September 16-18, 2019*. Ed. by M. Agosti, E. D. Buccio, M. Melucci, S. Mizzaro, G. Pasi, and F. Silvestri. Vol. 2441. CEUR Workshop Proceedings. CEUR-WS.org, 2019, pp. 21–24. URL: <https://ceur-ws.org/Vol-2441/paper23.pdf>.

Preprints

- [15] J. Augustine, **A. Cruciani**, and I. A. Gillani. *Highly Dynamic and Fully Distributed Data Structures*. 2025. arXiv: [2409.10235](https://arxiv.org/abs/2409.10235) [cs.DC]. URL: <https://arxiv.org/abs/2409.10235>.
- [16] **A. Cruciani**, A. Das, A. Raevskaya, and J. Suomela. *It does not matter how you define locally checkable labelings*. 2026. arXiv: [2602.18188](https://arxiv.org/abs/2602.18188) [cs.DC]. URL: <https://arxiv.org/abs/2602.18188>.
- [17] **A. Cruciani**, A. Das, M. Equi, H. Lievonon, D. Luong-Le, A. Modanese, and J. Suomela. *Is a LOCAL algorithm computable?* 2026. arXiv: [2602.21022](https://arxiv.org/abs/2602.21022) [cs.DC]. URL: <https://arxiv.org/abs/2602.21022>.
- [18] M. T. E. Abedesselam, **A. Cruciani**, F. Giacomelli, L. Kiffer, and F. Pasquale. *How Bitcoin Forms Its Network: Peer-Table Sampling and Structural Properties*. 2026.
- [19] J. Augustine, **A. Cruciani**, M. Kumar, S. Kuttan, and D. Peleg. *Gossip in an Anonymous Capacitated Networks*. 2026.

PhD Thesis

- [20] **A. Cruciani**. *Highly Dynamic and Fully Distributed Data Structures*. 2025. URL: <https://iris.gssi.it/handle/20.500.12571/34764>.

Presentations at Conferences & Workshops

◦ 2025:

- Maintaining a Bounded Degree Expander in Dynamic Peer-to-Peer Networks (SIROCCO, Durham, UK);

◦ 2025:

- Fast Percolation Centrality Approximation with Importance Sampling (ICDM, Washington DC, USA);
- Maintaining a Bounded Degree Expander in Dynamic Peer-To-Peer Networks (DISC, Berlin, DE);
- Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (DISC, Berlin, DE);
- Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (Joint Estonian-Latvian Theory Days, Riga, LV);
- Fast Percolation Centrality Approximation with Importance Sampling (Helsinki Algorithms & Theory Days, Helsinki, FI).

◦ 2024:

- Approximating Distance-based metrics through sampling (IIT Madras, Chennai, IN);
- Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (Online, Aalto University, Espoo, FI);
- Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (IIT Madras, Chennai, IN);
- On the Temporal Betweenness Centrality (IIT Madras, Chennai, IN);
- MANTRA: Temporal Betweenness Centrality Approximation through Sampling (ECML-PKDD, Vilnius, LT);
- Computing Distance-based metrics on Very Large Graphs (UNIPD, Padova, IT).

◦ 2023:

- PROPAGATE: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs (ECML-PKDD, Turin, IT);
- Proxying Betweenness Centrality Rankings in Temporal Networks (SEA, Barcelona, ES);
- Dynamic graph models inspired by the Bitcoin network-formation process (ICDCN, Kharagpur, IN).

◦ 2022:

- Dynamic graph models for the Bitcoin P2P network: simulation analysis for expansion and flooding time (SSS, Clermont-Ferrand, FR).

◦ 2019:

- About Graph Index Compression Techniques (IIR, Padova, IT);
- Iterative Compression technique for NP-Hard problems on Graphs (Uni. of Rome Tor Vergata, Rome, IT).

Additional Training

Bertinoro International Spring School

*Bertinoro, IT
Mar. 2022*

European Summer School on Learning in Games, Markets, and Online Decision Making

*Rome, IT
Sep. 2021*

Max Planck Advanced Course on the Foundations of Computer Science (Convex Optimization) (online)

*Saarbrücken, GE
Jul. - Aug. 2021*

Algorithmic Tools for Massive Network Analytics (online)

*Pisa, IT
May - Jun. 2021*

Max Planck Advanced Course on the Foundations of Computer Science (Market Design and Computational Fair Division) (online)

*Saarbrücken, GE
August 2020*

Algorithms and computational models for large-scale data analysis. University of Rome: “La Sapienza”. By Silvio Lattanzi (Google Research).

*Rome, IT
August 2019*

Academic Service

Program Committee

PC-Member: The 30th Conference on Principles of Distributed Systems (OPODIS 2026).

PC-Member: 28th International Conference on Distributed Computing and Networking (ICDCN 2027).

PC-Member: The 28th International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS 2026).

Reviewing

Conferences

Reviewer ICALP-PODC-DISC 2026

Reviewer STACS-PODC-WSDM-SDM-SEA-ISAAC-FSTTCS-ICANN 2025

Reviewer FUN 2020-2024

Reviewer AAMAS 2023

Journals

Reviewer: The Review of Socionetwork Strategies

Organization

Organizer of the workshop [Helsinki Theory Days 2026](#)

Students Supervision

Bachelor’s Students: Giacomo Rivetti (2025, co-supervised with Francesco Pasquale).

Master’s Students: Andrea Bonifati (2026, co-supervised with Francesco Pasquale).

Software Packages

- [PROPAGATE](#), an efficient algorithm for approximating various distance-based metrics (i.e., average distance, effective diameter, diameter and connectivity rate).
- [DREG](#), a dynamic expander graph generator.
- [TSBPROXY](#), a suite of efficient proxies for the temporal betweenness centrality rankings.
- [MANTRA](#), an efficient framework for approximating the temporal betweenness centrality using sampling.
- [FEPIC](#), an efficient approximation algorithm for the (doubly normalized) percolation centrality.
- [PERCIS](#), a fast approximation algorithm for the percolation centrality that uses importance sampling.

Programming Skills

- Basic: OWL, SPARQL, FORTRAN, COBOL, LISP
- Intermediate: GO, MATLAB, JAVASCRIPT, R, ASP.NET, PHP
- Advanced: PYTHON, JULIA, JAVA, C, C++, C#, SQL
- Frameworks: Apache Spark

Contact References

- John Augustine, IIT Madras: augustine@cse.iitm.ac.in
- Jukka Suomela, Aalto University: jukka.suomela@aalto.fi
- Francesco Pasquale, University of Rome Tor Vergata: pasquale@mat.uniroma2.it
- Leonardo Pellegrina, University of Padua: leonardo.pellegrina@unipd.it
- Giambattista Amati, Fondazione Ugo Bordoni: gba@fub.it
- Thorsten Götte, Hamburg University: thorsten.goette@uni-hamburg.de